

# StayMate

## Refined Product Requirement Document

*"Find your next stay, roommate, or rental space in minutes."*

**Target Audience:** Accommodation Seekers & Property Owners

**Architecture Update:** Monorepo, Contabo Infrastructure, In-App Chat, & OSM

# 1. Project Architecture & Setup

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## Monorepo Structure

The entire project will be hosted in a single unified Git repository (Monorepo) to share types, configurations, and streamline deployments. The structure includes:

- **/mobile**: React Native (Expo) application for Seekers and Owners.
- **/frontend**: React.js Web Application for the Admin Panel.
- **/backend**: Node.js/Express REST API & WebSockets.

## Infrastructure (Contabo)

- **Server Hosting**: Contabo VPS (Running Node.js backend & PostgreSQL DB).
- **Media Storage**: Contabo Object Storage (S3-Compatible Bucket) for property images and user avatars.

# 2. User Roles & Authentication

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## 2.1 Authentication (Email-Based)

The platform will primarily use Email authentication.

- **Registration/Login**: Email and Password (JWT based) or Email OTP.
- **Profile Information**: Full Name, Email, Gender, Occupation, Profile Photo.

## 2.2 User Roles

| Accommodation Seeker (Mobile App)                                                                                                                                                                           | Property Owner (Mobile App)                                                                                                                                        | Administrator (Web App)                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Search listings &amp; filter</li><li>• <b>Chat with owners directly</b></li><li>• <b>Chat with potential roommates</b></li><li>• Participate in community</li></ul> | <ul style="list-style-type: none"><li>• Manage properties &amp; media</li><li>• Set exact OSM location pin</li><li>• <b>Chat with interested seekers</b></li></ul> | <ul style="list-style-type: none"><li>• Manage users, properties &amp; chats</li><li>• Verify property owners</li><li>• Monitor platform (React Web Dashboard)</li></ul> |

## 3. Core Features

### Property Search & Maps (Open Source)

Replaces expensive Google Maps API with 100% free open-source solutions.

- **Map Data:** OpenStreetMap (OSM).
- **Map Renderer:** react-native-maps or MapLibre.
- **Geocoding:** Nominatim (for translating text to coordinates).
- **Functionality:** Users can search via radius (e.g., 5km), view embedded interactive maps, and owners can drop pins directly inside the app to save precise Latitude/Longitude.

## In-App Chat Module (Real-Time)

A built-in messaging system using Socket.io to keep users engaged inside the app.

- **Seeker to Owner:** Seekers can click "Message Owner" from a property listing.
- **Roommate Connect:** Users viewing the Community Polls or "Looking for Roommate" posts can initiate a direct chat.
- **Features:** Read receipts, message history, push notifications on new message.

## 4. Database Design (PostgreSQL)

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users: id, email, password_hash, name, mobile_number, gender, profile_image, role
properties: id, owner_id, title, description, property_type, rent_amount,
            latitude, longitude, address, district, city, created_at
property_images: id, property_id, image_url (Points to Contabo Bucket)
roommate_posts: id, user_id, title, budget, preferred_location, move_in_date
messages: id, sender_id, receiver_id, property_id (optional), content,
          is_read, created_at
favorites: id, user_id, property_id
notifications: id, user_id, title, message, status
```

## 5. Technology Stack

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| Domain              | Technology                               |
|---------------------|------------------------------------------|
| Mobile Application  | React Native, Expo, TypeScript           |
| Admin Web Dashboard | React.js, Tailwind CSS (/frontend)       |
| Backend Engine      | Node.js, Express.js (/backend)           |
| Database            | PostgreSQL                               |
| Real-Time Chat      | Socket.io (WebSockets)                   |
| Authentication      | JWT (JSON Web Tokens) via Email/Password |
| Location Services   | OpenStreetMap (OSM) & Nominatim          |
| Media Storage       | Contabo Object Storage (S3-Compatible)   |
| Server Hosting      | Contabo VPS                              |

## 6. Phased Rollout Plan

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### Phase 1: MVP Scope

✓ **Included:** Email Login, Property Listings, OpenStreetMap Radius Search, In-App Chat (User-Owner & User-User), Roommate Finder, Community Feed, Admin Web Dashboard, Contabo Infra.

✗ **Excluded:** Online Rent Payment, AI Recommendations, Paid Subscription Gateways.

## Phase 2 & Beyond

- Owner Verification System (Govt ID).
- Premium Listings & Advertisement integrations.
- AI-Based Room & Roommate Matching.
- Rent Payment Integration & Digital Agreements.